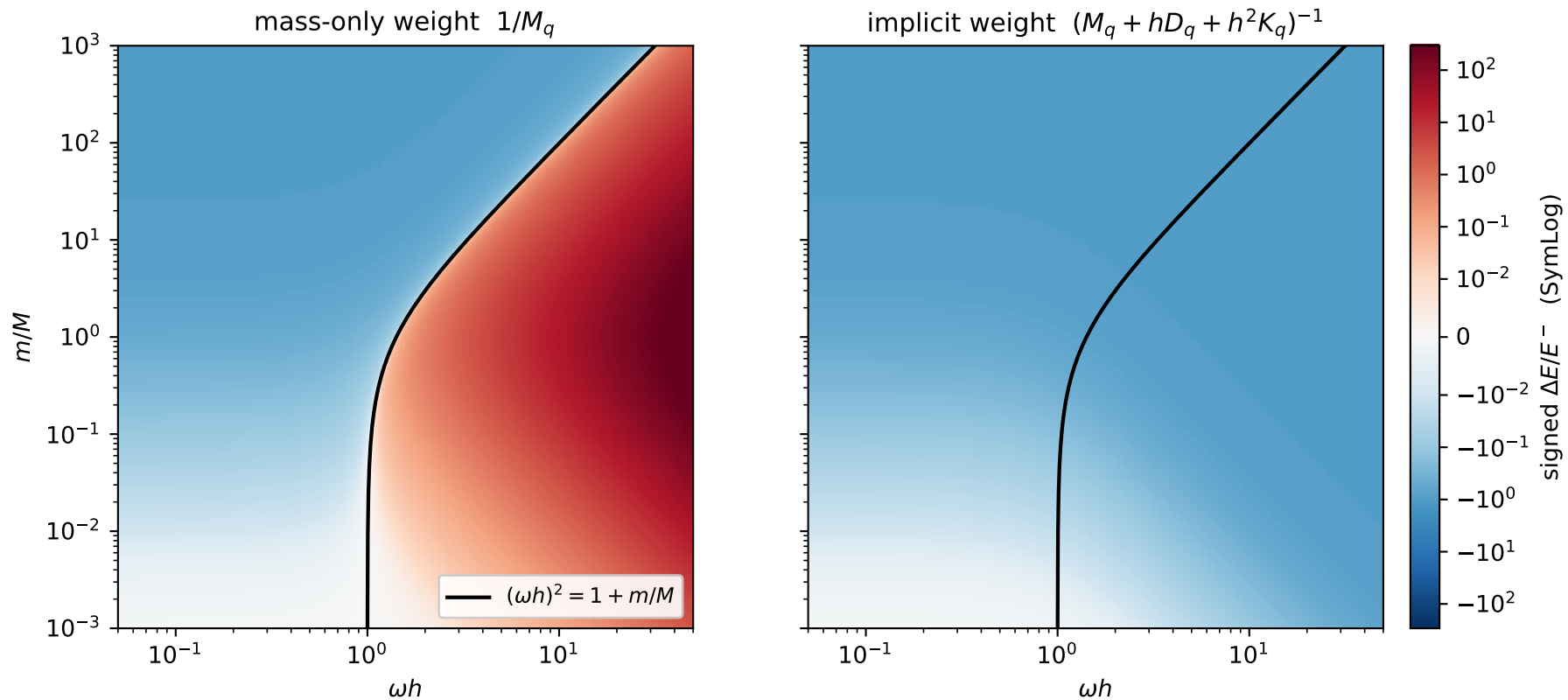


One-sweep injection phase map: measured sign vs analytic boundary



Left arm injects (red) exactly where the solid analytic curve predicts; the implicit arm is uniformly passive (blue). Hard contact. Hypotheses: one sweep, cold start, $e = 0$.